Cambridge International AS & A Level

BIOLOGY (9700) PAPER 2

Past Paper Questions By Topic
+ Answer Scheme

2015 - 2020

Complete Syllabus



Chapter 14

Biodiversity, classification and conservation

14.1 Biodiversity

244. 9700_s15_qp_22 Q: 6

(a) Sentences **(i)** and **(ii)** are extracted from longer definitions of ecological terms. Each has one or more missing words.

Complete **(i)** and **(ii)** using the correct terms chosen from the list below.

a population	**a community**	**an ecosystem**
a niche	**a habitat**	**trophic level**
producers	**organisms**	**consumers**

(i) ..... is the particular location and type of local environment occupied by ..... or organism, characterised by its physical features or by its dominant ..... . [3]

(ii) ..... is the functional role or place of a species of organism within ..... . [2]

(b) Only a small proportion of the light energy striking the leaves of producers becomes converted to chemical energy. The proportion that is converted is known as the photosynthetic efficiency (PE).

(i) Outline the possible reasons why a large proportion of light energy striking the leaves of plants is **not** converted into chemical energy.

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.....[3]

(ii) Most crops have a PE of 1% to 4%. Sugar cane, an important crop plant for food production and for the production of biofuel, has a PE of 7% to 8%.

Suggest the advantages of growing crops with high PE for food production or for biofuel.

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.....

.....[2]

(iii) Fertilisers containing nitrate are added to improve or maintain yield of crops such as sugar cane.

Name two organic compounds containing nitrogen that are made by plants and state one function of each in plant growth.

organic compound 1 .....

function .....

.....

organic compound 2 .....

function .....

.....

.....[2]

[Total: 12]

14.2 Conservation

245. 9700_s15_qp_23 Q: 6

In the oceans, parts of the nitrogen cycle involve different bacteria from those that are involved on the land.

A bacterium found in oceans is *Nitrococcus mobilis*, which carries out the following step in the nitrogen cycle:

$$NO_2^- \longrightarrow NO_3^-$$

nitrite nitrate

(a) (i) Name the stage in the nitrogen cycle in which this step occurs.

.....[1]

(ii) Describe how nitrogen in nitrate can be returned to the atmosphere in the form of nitrogen gas.

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.....[2]

(b) Phytoplankton are microscopic photosynthetic organisms that are the main producers in ocean ecosystems. Their habitat is the upper layers of the oceans where sunlight can penetrate through the water.

Define the terms:

(i) ecosystem

.....

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.....[2]

(ii) producer

.....

.....

.....[1]

(iii) habitat.

.....

.....

.....[1]

[Total: 7]